Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/22/2008
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/26/2000	**End Date:**	03/01/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: funding reinstated as a new grant, per D. Gillies (8/2006)		
Key Personnel Changes/Previous PI:	Dr. K. Lee and Dr. L. Zhang were employed to carry out the work. Ms. Jing Teng, a Ph. D. student carried out some NASA work		
COI Name (Institution):	Karma, A. (Northeastern University)		
Grant/Contract No.:	NAG8-1681		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed research will consist of accurate and rigorous numerical modeling of the dynamics of pattern evolution, coupled with critical ground-based and low gravity experimental studies in a model transparent system in which the dynamics of pattern evolution can be observed in situ. Initial ground-based studies have been carried out in the succinonitrile-salol and succinonitriole-water systems to establish the range of conditions over which cellular structures are stable under diffusive growth conditions. Cellular microstructures are complex since the interactions between the neighboring cells are significant and the steady state shape of the cell is a free boundary parameter. How the cellular structure reorganizes into a steady state microstructure is crucial to the understanding of the final microstructure that develops in materials. Specifically, the understanding of the physics that gives rise to multiple stable cell configurations under a given growth condition is critical from a scientific viewpoint. We have, thus, focused a part of the research onto the physics behind the multiple solution of cells in an array and we have investigated the self-similarity of cell shapes and compared them with analogous scaling laws that are observed in different fields of sciences. The ability to design microstructures of solidified materials is crucial for obtaining optimum properties of materials processed by casting, welding, or rapid solidification techniques. It has been well-established that the microstructure selection occurs during the dynamic growth conditions. The objective of the task is to establish fundamental principles that govern the spatio-temporal evolution of cellular and dendritic interface patterns in directional solidification. The results will be applied to other analogous patterns that form in different disciplines of sciences. Low gravity experiments will be used to validate a rigorous numerical model of pattern evolution dynamics that we have now developed by using the phase-field approach. The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict reliable processing conditions that optimize microstructures and hence material properties and performance. In addition, the knowledge gained from this study will be applied to other pattern forming phenomena that occur in different fields of science such as combustion and fluid physics. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An experimental set-up was designed to carry out directional solidification of transparent materials in capillary samples in which convection effects are minimized. This equipment can examine small scale microstructures terrestrially under diffusive growth conditions. The results of our experiments have established some critical physics on the role of experimental variables and physical properties of single cell and single dendrites which are then validated by a detailed numerical modeling. These fundamental results are critical in the modeling of casting and welding microstructures that are now used by industry to design appropriate processing conditions. These studies have also been correlated with microstructures in other disciplines of sciences including some biological systems. These results are extensively used by scientists in different areas of sciences.
Task Progress:	The following aspects of solidification were examined. (i) Determination of the appropriate system for low gravity experiments in space. We considered succinonitrile base with different solutes. The solute we examined was: acetone, salol, water and camphor. We finally selected succinonitrile-camphor system since the other three systems had problems for space experiments. (ii) We carried out detailed studies in Succinonitrile-salol and water system that included the characterization of processing conditions for planar, cellular and dendritic interfaces so that experimental matrix can be defined for space experiments. (iii) The succinonitrile-camphor system was first characterized. Refractive index was measured as a function of composition and temperature since it is required for the proper design of the lenses in the directional solidification insert in DECLIC. (iv) The transition from cellular to dendritic interface was characterized quantitatively. We observed that this transition is not sharp but occurs locally over a range of velocity or thermal gradient. This was attributed to the presence of a finite range of spacing in a given cellular array. We characterized the condition for the onset of the transition, and obtained an expression for the critical cell spacing at which sidewise perturbations become stable. This critical spacing was determined as a function of velocity, composition and thermal gradient. It was found that the critical spacing is given by the geometrical mean of the thermal, solutal and capillary lengths. (v) Detailed study of the dynamics of interface pattern evolution was examined in the Succinonitrile-water system. These results were compared with the observation in space obtained from the glove box experiments. (vi) The effect of convection was examined in the bulk sample. It was found that significant irregularities in the pattern formation occur due to the presence of convection. (vii) A thin tube technique was developed and used to quantitatively examine the effect of convection on microstructure. By using a range of sample diameters (0.2 mm to 5.5 mm), the effect of increasing convection on solidification microstructure was quantitatively established. The capillary sample technique was then applied to measure the diffusion coefficient. It was shown that the value of the measured diffusion coefficient increased as the sample diameter was increased. The true value of the diffusion coefficient was obtained in very thin samples of diameters 0.4 and 0.6 mm. The thin tube technique was also used to characterize quantitatively the effect of convection on eutectic microstructure. (viii) A detailed theoretical model of the effect of convection on microstructure was developed, and it was applied to the novel microstructures that we observed in peritectic system. In presence of convection one forms a large treelike structure of the primary phase that is surrounded by the peritectic phase. Such structures are not predicted by the diffusive growth model, and out theoretical model showed the mechanism of formation of these structures that is governed by the presence of oscillatory convection in the liquid. Detailed experiments in diffusive and convective growth conditions were carried out to compare the microstructure changes that occur as the convection becomes

	important. (ix) A phase-field model was developed to examine the dynamics of interface pattern evolution in two-dimensions. The effect the anisotropy in interface energy on cellular microstructure was examined and it was shown that increasing anisotropy alters the cell spacing, cell tip radius and cell tip undercooling. This model will be extended to the three-dimensional structures.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
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